

MECH3 Formula Sheet

Plane Stresses

Analytical Mohr's Circle Calculations:

$$\begin{aligned}\sigma_P, \sigma_Q &= C_x \pm R \\ &= \frac{\sigma_x + \sigma_y}{2} \pm \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2} \\ \tau_{max} &= R \\ &= \pm \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2} \text{ [NOT GIVEN]}\end{aligned}$$

Rankine's Yield Criterion for Brittle Materials

$$\sigma_P = \sigma_w = \frac{\sigma_Y}{n} \text{ [NOT GIVEN]}$$

Tresca's Yield Criterion for Ductile Materials

$$\tau_{max} = \frac{\sigma_w}{2} = \frac{\sigma_Y}{2n} \text{ [NOT GIVEN]}$$

Plane Strains

Poisson's Ratio

$$\nu = -\frac{\varepsilon_y}{\varepsilon_x}, \text{ when loaded axially about } x \text{ [NOT GIVEN]}$$

Dimensional Changes

$$\begin{aligned}\Delta x &= \varepsilon_x x_0 \text{ [NOT GIVEN]} \\ \Delta A &\approx (\varepsilon_x + \varepsilon_y) A_0 \text{ [NOT GIVEN]} \\ \Delta V &\approx (\varepsilon_x + \varepsilon_y + \varepsilon_z) V_0\end{aligned}$$

Rectangular Rosette Formulae (for counter clockwise flowing gauges)

$$\begin{aligned}C_x &= \frac{\varepsilon_1 + \varepsilon_3}{2} \text{ [NOT GIVEN]} \\ \phi_1 &= -2\varepsilon_2 + \varepsilon_1 + \varepsilon_3 \text{ [NOT GIVEN]}\end{aligned}$$

Delta Rosette Formulae (for counter clockwise flowing gauges)

$$C_x = \frac{1}{2} \left(\varepsilon_1 + \frac{2(\varepsilon_2 + \varepsilon_3) - \varepsilon_1}{3} \right) \text{ [NOT GIVEN]}$$

$$\phi_1 = -\frac{2}{\sqrt{3}}(\varepsilon_3 - \varepsilon_2) \text{ [NOT GIVEN]}$$

Star Rosette Formulae (for counter clockwise flowing gauges)

$$C_x = \frac{1}{2} \left(\varepsilon_1 + \frac{2(\varepsilon_2 + \varepsilon_3) - \varepsilon_1}{3} \right) \text{ [NOT GIVEN]}$$

$$\phi_1 = -\frac{2}{\sqrt{3}}(\varepsilon_3 - \varepsilon_2) \text{ [NOT GIVEN]}$$

Stress to Strain Equations

$$\varepsilon_x = \frac{1}{E}(\sigma_x - \nu\sigma_y)$$

$$\varepsilon_z = \frac{1}{E}(-\nu\sigma_x - \nu\sigma_y)$$

Strain to Stress Equations

$$\sigma_x = \frac{E}{1 - \nu^2}(\varepsilon_x + \nu\varepsilon_y)$$

Shaft Applications

Axial Stress

$$\sigma_A = \frac{F}{A} = \frac{F}{\frac{\pi}{4}(D^2 - d^2)} \text{ [NOT GIVEN]}$$

Bending Stress

$$(\max) \sigma_B = \pm \frac{My}{I} = \pm \frac{M \left(\frac{D}{2} \right)}{\frac{\pi}{64}(D^4 - d^4)}$$

Torsion Stress

$$\tau_T = \frac{Tr}{J} = \frac{T \left(\frac{D}{2} \right)}{\frac{\pi}{32}(D^4 - d^4)}$$

Pressure Vessels

Thin-Wall Criterion

$$\frac{t}{d} < 0.05 \text{ [NOT GIVEN]}$$

Hoop Stress in a Spherical shell

$$\sigma_H = \frac{P}{4 \left(\frac{t}{d} \right)}$$

Dimensional Change in Diameter & Circumference

$$\begin{aligned}\Delta d &= \varepsilon_H d_0 \text{ [NOT GIVEN]} \\ \Delta C &= \varepsilon_H C_0 = \varepsilon_H (2\pi d_0) \text{ [NOT GIVEN]}\end{aligned}$$

Dimensional Change in Wall Thickness

$$\begin{aligned}\varepsilon_t &= \frac{1}{E}(-2\nu\sigma_H) \Rightarrow \Delta t = \varepsilon_t t_0 \text{ (sphere) [NOT GIVEN]} \\ \varepsilon_t &= \frac{1}{E}(-\nu\sigma_H - \nu\sigma_L) \Rightarrow \Delta t = \varepsilon_t t_0 \text{ (sphere) [NOT GIVEN]}\end{aligned}$$

Dimensional Change in Cross-sectional area and Volume for Spherical Vessels

$$\begin{aligned}\Delta A &= 2\varepsilon_H A_0 = 2\varepsilon_H \frac{\pi d_0^2}{4} \text{ [NOT GIVEN]} \\ \Delta V &= 3\varepsilon_H V_0 = 3\varepsilon_H \frac{\pi d_0^3}{6} \text{ [NOT GIVEN]}\end{aligned}$$

Hoop Stress in a Cylindrical Shell

$$\sigma_H = \frac{P}{2 \left(\frac{t}{d} \right)}$$

Longitudinal Stress in a Cylindrical Shell

$$\begin{aligned}\sigma_L &= \frac{P}{4 \left(\frac{t}{d} \right)} \text{ (CLOSED ENDS) [NOT GIVEN]} \\ \sigma_L &= 0 \text{ (OPEN ENDS) [NOT GIVEN]}\end{aligned}$$

General Equilibrium Equation for Hoop Stress with Internal Pressure

$$\begin{aligned}\sum F_L &= 0 \\ F_{\text{applied}} + F_{\text{internal}} - F_{\text{pressure}} &= 0 \\ F + \sigma_L \pi d \times t - P \frac{\pi d^2}{4} &= 0 \text{ [NOT GIVEN]}\end{aligned}$$

Bulk Modulus of a Fluid

$$K = \frac{P}{-\frac{\Delta V}{V_0}} \Rightarrow \Delta V = -\frac{PV_0}{K}$$

Short Columns

Axial & Bending Stresses of Rectangular Columns

$$\sigma_A = \frac{F}{A} = \frac{F}{bd}$$

$$(\max) \sigma_B = \frac{My}{I} = \frac{M \frac{d}{2}}{\frac{bd^3}{12}}$$

Tension developed in q rows of n bolts

$$M = q \sum k F_t * d_i$$

$$M = q \sum_{i=1}^n \frac{i-1}{n-1} F_t \times (i-1)d$$

$$M = q \left(\frac{1}{n-1} F_t d + \frac{2}{n-1} F_t (2d) + \frac{3}{n-1} F_t (3d) + \dots + \frac{n-1}{n-1} F_t (n-1)d \right)$$

$$M = q F_t d \left(\frac{1}{n-1} + \frac{4}{n-1} + \frac{9}{n-1} + \dots + (n-1) \right)$$

$$M = q F_t d \left(\frac{n(2n-1)}{6} \right)$$